MANIKONDA THARUN

Software Engineer

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SUMMARY

Software engineer with 3+ years of experience building distributed backend services, high-throughput data pipelines, transaction validation systems, and auditable enterprise workflows. Develops Python, Go, and Java services with Kafka, PostgreSQL, Redis, Cassandra, Docker, Kubernetes, AWS, and GCP. Experienced in idempotent processing, retries, caching, data validation, observability, automated testing, and cross-functional delivery across product, platform, compliance, and operations teams.

TECHNICAL SKILLS

Languages: Python, Go, Java, SQL, JavaScript
Backend and Distributed Systems: REST APIs, gRPC, Spring Boot, FastAPI, Microservices, Event-Driven Architecture
Data and Messaging: Kafka, PostgreSQL, Cassandra, Redis, Data Pipelines, Data Validation, Caching, SQL Data Modeling
Correctness and Reliability: Idempotency, Retry Handling, Data Consistency, Audit Logging, Automated Testing, Load Testing
Cloud and Platform: AWS, GCP, Docker, Kubernetes, Terraform, GitHub Actions, Linux
Observability: Prometheus, Grafana, OpenTelemetry, Structured Logging, Metrics and Alerting

PROFESSIONAL EXPERIENCE

McKinsey & Company | CA, USA
Software Engineer May 2025 - Present

- Built Python and Go ingestion services that processed approximately 10 million records monthly, reducing end-to-end data latency by 40% while preserving validation and downstream consistency.
- Developed REST and gRPC microservices with PostgreSQL and Redis on Kubernetes, lowering service error rates by 35% and supporting horizontal scaling to approximately 50 pods.
- Implemented idempotent request handling, retry policies, schema validation, and failure isolation for high-concurrency workflows, improving predictable recovery from partial processing failures.
- Designed metadata-rich retrieval workflows and operational logs that made data movement traceable across ingestion, indexing, and serving stages for engineering and audit reviews.
- Instrumented distributed services with Prometheus, Grafana, and OpenTelemetry, reducing mean time to detect incidents by 45% across production data workflows.
- Partnered with product, data, and platform teams to define acceptance criteria, document API behavior, and deliver Terraform and GitHub Actions changes through controlled releases.

Uber | CA, USA
Backend Engineer February 2024 - May 2025

- Built Kafka and Java transaction-validation pipelines for more than 75,000 daily bookings, increasing peak throughput by 50% while maintaining deterministic processing outcomes.
- Developed Go and Spring Boot gRPC services on Kubernetes with TLS and AWS IAM, reducing inter-service latency by 27% while enforcing authenticated service access.
- Implemented Redis and Cassandra caching strategies that reduced downstream database calls by 60% and improved p95 response time by approximately 35%.
- Applied idempotency keys, retry controls, and event-level validation to separate recoverable failures from permanent exceptions without interrupting routine transaction processing.
- Added k6 load tests, automated integration checks, and GitHub Actions quality gates, identifying performance regressions before release and improving deployment stability by 40%.
- Built Prometheus and Grafana dashboards for transaction failures, latency, and recovery workflows, lowering mean time to recovery by 30% during operational incidents.

KPMG | India
Java Backend Engineer September 2021 - July 2022

- Developed Java and Spring Boot services for audit and compliance workflows, improving document-processing reliability across 15 downstream reporting processes.
- Built PostgreSQL-backed REST APIs with validation and traceable status transitions, reducing report-generation latency by 30% while preserving consistent audit records.
- Implemented Kafka consumers with ordering and idempotency controls for approximately 120,000 monthly documents, reducing duplicate records by 40%.
- Created JUnit integration tests and Docker-based CI checks for document ingestion and exception handling, catching 15 critical regressions before release.
- Added AWS-hosted Prometheus and Grafana monitoring for 10 core services, reducing mean time to detect processing failures by 35% and supporting SLA reporting.

Trigent Software | India
Frontend Engineer March 2020 - August 2021

- Delivered responsive React interfaces for enterprise workflows and integrated REST APIs with authenticated session handling across eight application modules.

- Created Jest and integration tests that detected more than 15 UI regressions before release and improved reliability across shared feature branches.

PROJECTS

Enterprise AI Document Assistant

- Built a Python and LangChain retrieval pipeline for more than 5,000 enterprise documents with 92% retrieval precision and semantic-search responses under 500 ms.
- Implemented FastAPI endpoints, Pinecone indexing, AWS S3 storage, and automated validation workflows for repeatable ingestion and retrieval behavior.

NotchSafe macOS Productivity Utility

- Engineered a Swift and SwiftUI utility for encrypted local storage and clipboard management using AES-256-GCM and Touch ID authentication.
- Designed persistent event monitoring while maintaining a binary under 5 MB and low system memory usage.

EDUCATION

Master of Science in Computer Science
University of Alabama at Birmingham — Aug 2022 - Dec 2023

Bachelor of Engineering in Computer Science
K.S. Institute of Technology, Karnataka, India — Aug 2018 - May 2022

CERTIFICATION

AWS Certified Solutions Architect - Associate